

15 Diseases of Mango

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Introduction

Mango, *Mangifera indica* (family: *Anacardiaceae*), is the world's fifth most important fruit crop (FAO, 2000). Global production in 2000 was 24.8 million tonnes (Mt), 12 Mt of which were harvested in India. Other major producers included, in descending order, China, Mexico, Thailand, Pakistan, Philippines, Indonesia, Nigeria and Brazil. The overwhelming majority of these fruit are consumed in the producing countries. In 1999, only 2% of the total harvest was exported as fresh fruit (Mexico was the leading exporter with 204,002 t), and <0.1% was exported as juice or pulp.

M. indica is one of 69 species in the genus, at least 27 of which produce edible fruit (Mukherjee, 1997). Although the species name suggests that it originated in India, it is now thought to have evolved in a large area that includes present day Bangladesh, northeastern India and northwestern Mynamar (Burma). Mango is highly adaptable, and is grown between ~35° N and S latitudes in a wide range of soils and environmental conditions.

Within the species, two types are recognized. Monoembryonic mangoes originated in subtropical India and produce a seed with a single zygotic embryo. In contrast, polyembryonic mangoes arose in Southeast Asia and produce a seed with several embryos, one of which is zygotic and the others which are

usually nucellar. Nucellar embryos of the polyembryonic 'Turpentine' and '13-1' are used for rootstocks in Florida and Israel, and those of 'Kensington Pride' are used to propagate this cultivar in Australia (Crane *et al.*, 1997).

Hundreds of mango cultivars exist. Although superior types have been selected for several centuries in India, organized introductions of parents and the evaluation of their progeny has occurred only during the last century (Knight, 1997). In 1889, grafted cultivars were first introduced from India to Florida. During the ensuing decades, superior individuals from several generations of open-pollinated progeny were selected and brought into production. The Florida cultivars possess desirable traits such as uniform bearing, reddish skin coloration and tolerance to anthracnose. Many of these cultivars, most notably 'Tommy Atkins', are now used worldwide.

The introduction of superior parents and evaluation of their seedling progeny now occurs in several countries (Knight, 1997). In virtually all of these programmes, the male, pollen parent is not known. Mango has a long juvenile period, and fruit retention after pollination is very low. Most parents are heterogeneous and only one zygotic embryo is produced per fruit. Thus, evaluation of large seedling populations is a major undertaking. Classical breeding has produced some useful

hybrids, but open pollinated germplasm is far more important (Iyer and Degani, 1997).

The Major Diseases

Diseases affect every organ of this important tree (Ploetz and Prakash, 1997). They damage seedlings and grafted plants in the nursery, decrease fruit set and retention, and are important pre- and postharvest problems on fruit. In many areas, they are the most important constraint to fruit production. Although most of these diseases are caused by fungi, an alga and bacterium are also important pathogens. No mango disease is known to be caused by phytoplasma, protozoa, viroid or virus agents.

Algal leaf spot (red rust)

Algal leaf spot, which is also known as red rust, is a common problem in the tropics and subtropics (Joubert and Rijkenberg, 1971). On mango, the disease is usually serious only in poorly managed orchards (Lim and Khoo, 1985). In these situations, mites, insects and other foliar diseases can increase the severity of the disease.

Algal leaf spot is caused by the green alga, *Cephaleuros virescens* (Lim and Khoo, 1985). It is described in Chapter 1, as are the symptoms, epidemiology and management of the disease.

Alternaria rot (black spot)

Alternaria rot, which is also known as black spot, causes postharvest fruit rot in Australia, Egypt, India, Israel and South Africa. Most commercial cultivars are susceptible. The pathogen also affects leaves and panicles; the latter problem is covered below under Blossom blight.

Symptoms

On leaves, symptoms are present throughout the year as round, black spots that are 1–3 mm in diameter and most noticeable on the

underside of leaves (Prusky, 1994). Similar lesions develop around lenticels on fruit (Fig. 15.1). They usually start near the stem end, but can expand and merge to cover much of the fruit surface. Affected areas do not soften or penetrate more than 1–2 mm into the flesh until late in symptom development. Lesion centres are depressed and develop olive-brown conidia of the pathogen under moist conditions. Symptoms of Alternaria rot are more restricted, and affected tissue is darker and harder than that affected by anthracnose (Plate 85).

Causal agent

Alternaria alternata is described in Chapter 1 (Prusky *et al.*, 1983; Cronje *et al.*, 1990). The fungus is widespread and has a large host range (Neergaard, 1945; Domsch *et al.*, 1980). Diseases that it causes on mango are generally significant only in arid environments. For example, in Israel, it is the most important fruit disease (Prusky, 1994).

Epidemiology

Infected leaves, twigs and inflorescences are significant sources of inoculum for fruit infection, as are leaves on the orchard floor. Conidia of *A. alternata* are moved by air currents and in dew runoff (Prusky, 1994).

Fruit infection has been related to the length of time over 350 h that relative humidity is >80%. These infections develop after ripening begins (Prusky *et al.*, 1983). Fungitoxic levels of two resorcinol compounds have been associated with infection latency (Droby *et al.*, 1986, 1987). They are present in the peel prior to ripening, and as ripening begins their concentrations decline and postharvest decay begins.

Management

'Keitt' is more resistant than 'Haden' and 'Tommy Atkins' (Droby *et al.*, 1986). Losses can be minimized by regular fungicide applications in the field, postharvest fungicide treatment and enhanced ripening with ethylene (Prusky, 1994). Three applications of the fungicide maneb beginning 2 weeks after



Fig. 15.1. Initial symptoms of *Alternaria* rot that are associated with lenticels on the surface of a mango fruit (photo: D. Prusky).

fruit set were most effective before harvest, and efficacy decreased if treatment was delayed. Postharvest treatment with prochloraz was as effective as three preharvest treatments. A brush and hot water (50–60°C) treatment was shown to be as effective as postharvest treatment with prochloraz 3 weeks after storage (Prusky *et al.*, 1997).

Anthracnose

Anthracnose is the most important disease of mango in all but arid production areas (Cook, 1975; Lim and Khoo, 1985; Dodd *et al.*, 1997; Ploetz and Prakash, 1997). Because it causes unsightly blemishes on fruit, it is a

major pre- and postharvest problem. Anthracnose can also damage foliage, and under crowded and moist conditions causes serious problems in nurseries and young orchards (Bose *et al.*, 1973). The damage it causes on panicles is covered later in this chapter under Blossom blight.

Symptoms

New leaves that emerge during rainy periods are most susceptible (Fitzell and Peak, 1984; Jeffries *et al.*, 1990). Leaf lesions begin as small, dark brown spots that are surrounded by chlorotic haloes, have irregular margins and are not limited by leaf veins. Leaf lesions usually remain small, but under

warm and humid conditions enlarge and coalesce to form irregular patches, ≥ 1 cm in diameter (Plate 86). Centres of old lesions deteriorate and fall from the leaf, resulting in a perforated, tattered appearance.

Anthrachnose also causes twig dieback, although it is a less common cause of this symptom than other fungi (Lim and Khoo, 1985; Ploetz *et al.*, 1996a). In severe cases, elongated, blackened lesions form on twigs that dry from the tip.

Small fruit can develop minute brown spots and abort if infected early in their development. After fruit exceed 4–5 cm in diameter, abortion is less common and infections tend to stop development once an appressorium is formed. Further development occurs after fruit mature and begin to ripen. In these cases, irregular, dark brown to black lesions develop that are somewhat depressed and can crack the fruit surface (Fig. 15.2). Under humid conditions, large areas may be involved, and orange to pinkish masses of conidia are formed on the decaying surface. Lesions can form anywhere on fruit, but lin-

ear smears that radiate from the stem end to the apex are common (Plate 87). Lesions on fruit initially are superficial, and penetrate deeper than 5 mm into the mesocarp only during the final stages of development.

Causal agents

Anthrachnose is caused by three closely related taxa of fungi. In most regions, *Colletotrichum gloeosporioides* (teleomorph: *Glomerella cingulata*) is the responsible agent (Dodd *et al.*, 1997). In Australia, a variant of this fungus that produces smaller conidia, *C. gloeosporioides* var. *minor*, is involved, and in Australia and India, *C. acutatum* (teleomorph: *Glomerella acutata*) plays a minor role (Fitzell, 1979; Prakash, 1990). *C. gloeosporioides* and *C. acutatum* are described in Chapter 1.

Gantotti and Davis (1993) reported that isolates of *C. gloeosporioides* from different mango organs produced different pectin-degrading enzymes. Ten different pectic zymogram groups (PZGs) were identified, based on the array of enzymes that the iso-

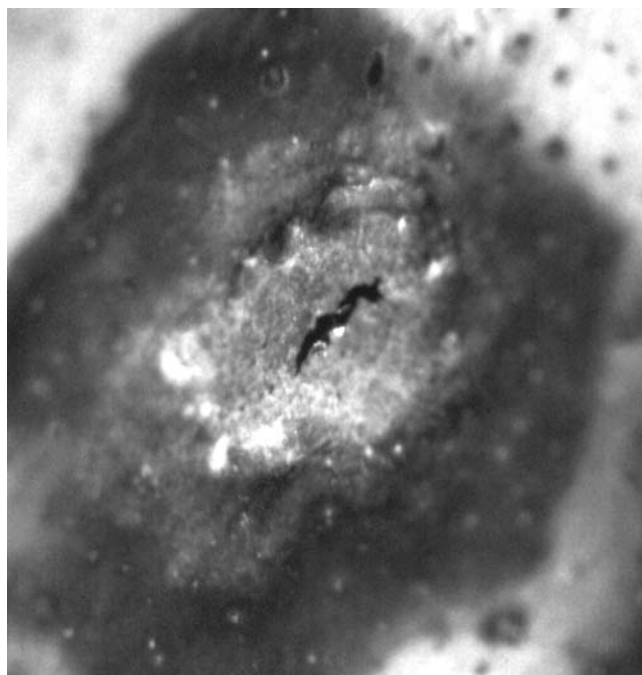


Fig. 15.2. An anthrachnose lesion on the surface of mango fruit that has cracked and become covered with mycelia and conidia of the causal fungus, *Colletotrichum gloeosporioides* (photo: R.C. Ploetz).

lates produced. The authors indicated that the fungus on mango was genetically diverse, and suggested that isolates from some of the PZGs were adapted to different organs. In contrast, Alahakoon *et al.* (1994b) indicated that >80% of the isolates that were isolated from mango leaves, inflorescences and fruit in Sri Lanka were identical, based on restriction fragment length polymorphisms (RFLPs) of genomic DNA.

Related work indicated that isolates from mango may comprise a pathogenically and genetically distinct population of *C. gloeosporioides* (Hodson *et al.*, 1993; Alahakoon *et al.*, 1994a; Hayden *et al.*, 1994). Although most of the isolates that were tested were collected in Australia and Sri Lanka, isolates from other countries were also represented. Analyses of RFLPs and random amplified polymorphic DNAs (RAPDs) indicated that isolates from mango were genetically uniform and differed from those that were recovered from avocado, carambola and papaya. Isolates that were genetically similar to those from the latter fruits were found infrequently on mango. The mango isolates were not found on the other crops and usually were virulent only on mango. Alahakoon *et al.* (1994b) suggested that the mango isolates comprised a population of *C. gloeosporioides* that was disseminated throughout the world from a single source, perhaps as an endophyte.

Epidemiology

Fitzell and Peak (1984) determined that conidia were the most important type of inoculum in Australia. They were produced on branch terminals, mummified inflorescences, flower bracts and leaves. New leaf flushes were the most significant source of inoculum, a finding that was corroborated by Dodd *et al.* (1991) in the Philippines. Optimum production of conidia occurred between 25 and 30°C when free moisture was available, but also formed at relative humidities of 95–97%. The pathogen's teleomorph played no apparent role in the spread of the disease (Fitzell and Peak, 1984).

Conidia usually are dispersed by rain-splash, and infection requires free moisture

(Jeffries *et al.*, 1990). In water, conidia form germ tubes within 6–8 h, and appressoria within 10–12 h. Germ tubes are usually <20 µm long and terminate in appressoria. As they age, appressoria become melanized. This attribute strengthens the appressorium and facilitates penetration of the cuticle by infection pegs that it produces. Infection is direct.

Management

Although some mango cultivars are moderately tolerant, none are sufficiently resistant to be produced without fungicides in humid areas (Dodd *et al.*, 1997). Several systemic and non-systemic fungicides are effective, and local authorities should be consulted regarding their registration and use. Care should be exercised when one of these, chlorothalonil, is used since it is phytotoxic when applied to fruit that are larger than a golf ball (Fig. 15.3).

Fungicide application focuses on reducing damage to fruit and inflorescences. Since infected foliage and branches are important sources of inoculum, fruit set and anthracnose control on fruit are enhanced if disease control is exercised prior to flowering (Jeffries *et al.*, 1990). Trees may also require protection in crowded nurseries that receive overhead irrigation.

Dodd *et al.* (1991) developed a model that was based on key criteria for infection. It was used to reduce the number of applications of a systemic fungicide to control anthracnose on fruit.

Although fruit-to-fruit spread of anthracnose after harvest is unlikely, postharvest control of latent infections often is needed, particularly if fruit are stored or shipped (Dodd *et al.*, 1997). Hot water treatments and fungicide dips are most common. Cultivars vary in their tolerance to hot water, and treatment temperatures should never exceed 55°C for 5 min. When hot-water treatment is combined with fungicides (e.g. benomyl or imazalil), water temperatures should be reduced to 52 or 53°C. Cold fungicide dips are less effective. Gamma-ray irradiation has been tested, but is not very effective at the doses that fruit of most cultivars tolerate (Johnson *et al.*, 1997).



Fig. 15.3. Cracks and fissures on the surface of 'Keitt' mango fruit caused by applications of chlorothalonil. Although this fungicide is effective against blossom blight and anthracnose, it should not be applied when fruit are larger than a golf ball (photo: R.C. Ploetz).

Bacterial black spot (black canker)

Bacterial black spot is a major constraint in many production areas, and fruit losses of >50% can occur on the most susceptible cultivars. It can be the most important disease where fungal-induced diseases are well controlled (Gagnevin and Pruvost, 2001). In India, the disease is called bacterial canker due to the cankers it causes on stems of susceptible cultivars (Prakash *et al.*, 1994). Bacterial black spot has been identified in Australia, the Comoros, India, Japan, Kenya, Malaysia, Mauritius, New Caledonia, Pakistan, the Philippines, Réunion, Rodrigues, South Africa, Taiwan, Thailand and the United Arab Emirates (Fukuda *et al.*, 1990; Pruvost *et al.*, 1992; Prakash *et al.*, 1994; Gagnevin and Pruvost, 1995, 2001; Kishun, 1995). The legitimacy of the only report from the western hemisphere (Brazil) has been questioned (Gagnevin and Pruvost, 2001).

Symptoms

Mango leaves, stems and fruit are affected (Manicom and Pruvost, 1994; Gagnevin and Pruvost, 2001). On leaves, water-soaked spots are initially 1–3 mm in diameter. As they enlarge, they become raised, black and angular, are limited by veins and surrounded by chlorotic haloes (Fig. 15.4A). These lesions are larger and more conspicuously raised than those caused by bacterial spot (Fig. 15.4B). Lesions can merge to form large necrotic patches, and bacteria may ooze from lesions during wet conditions. Old lesions dry out, turn white or grey, and crack. Defoliation occurs in severe cases. Anthracnose lesions are not raised or as black and angular as those that are caused by bacterial black spot.

Fruit lesions start as water-soaked haloes around lenticels and wounds. They become raised, blacken, crack along the long axis, and range from 1 to 15 mm in diameter. They are irregular in shape, can extend 8–15 mm

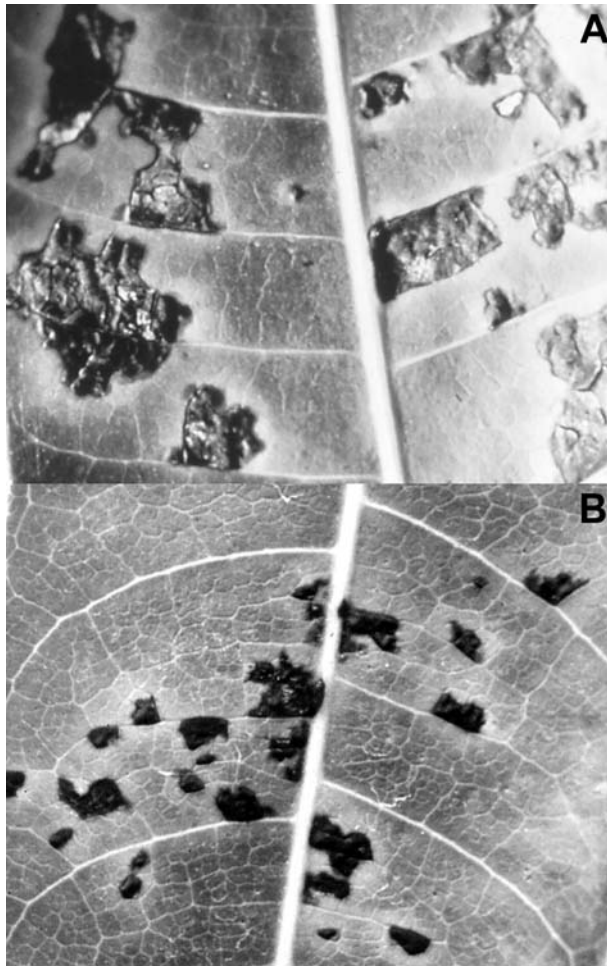


Fig. 15.4. (A) Symptoms of bacterial black spot on the undersurface of a mango leaf, caused by *Xanthomonas* sp. pv. *mangiferaeindicae* (photo: O. Pruvost) and (B) symptoms of bacterial spot on the undersurface of a mango leaf, caused by a yellow pigmented xanthomonad (photo: R.C. Ploetz). Bacterial black spot lesions are larger and more raised than those of bacterial spot.

into the mesocarp, and usually produce a gummy exudate (Plate 88). Young fruit abscise when they or panicles are affected.

On branches, bacterial black spot lesions are dark and cracked along the long axis (Plate 89). They develop only on highly susceptible cultivars and often are associated with wounds.

Causal agent

Xanthomonas campestris pv. *mangiferaeindicae* is the most commonly used epithet for this

pathogen. However, Gagnevin and Pruvost (2001) noted that the pathogen had not been included in recent taxonomic studies. Until it was examined with current methods, they felt that the provisional name *X. sp. pv. mangiferaeindicae* was more appropriate.

Early reports of *Pseudomonas mangiferaeindicae* and *Erwinia mangiferae* as incitants were probably in error (Patel *et al.*, 1948; Steyn *et al.*, 1974). The bacterium's placement in *Pseudomonas* was probably due to it producing non-pigmented colonies in culture, and Cook (1975) indicated that *E. mangiferae*

was a saprophyte that reached high populations in old lesions. *Pseudomonas syringae* pv. *syringae* causes a different disease of mango, apical necrosis (Cazorla *et al.*, 1998).

The pathogen has one flagellum, is Gram-negative, rod-shaped and $0.4\text{--}0.5 \times 1.0\text{--}1.5$ μm (Manicom and Wallis, 1984). It is oxidase negative and does not reduce nitrates to nitrites. It cannot use asparagine as a sole carbon and nitrogen source, but is able to hydrolyse starch, esculin, gelatin and casein. On artificial media, colonies are cream coloured. (The latter trait is atypical for xanthomonads, which usually are yellow in culture.) Yellow-pigmented xanthomonads have been recovered from mango leaves in Brazil, Réunion, South Africa and the USA (Florida). They are weak pathogens (see Fig. 15.4B) that do not affect fruit and differ in several criteria from *bona fide* strains of the pathogen (M.J. Davis, personal communication). Since these strains can confuse diagnosis, reliable detection of the disease relies on pathogenicity tests and identification of the pathogen. In the latter regard, a semi-selective medium, mono- and polyclonal antibodies, and RFLP analysis of the *hrp* gene cluster have been used (Gagnevin and Pruvost, 2001).

Epidemiology

The pathogen is disseminated by wind-driven rain (Gagnevin and Pruvost, 2001). Long-distance dissemination occurs via infected propagation material and, less frequently, in infected fruit pits. True seed are not infected, but surface contamination is known. Insects may play a role in dissemination, but these interactions are little studied.

The pathogen is an epiphytic colonist of leaves (Manicom, 1986; Pruvost *et al.*, 1990), buds (Pruvost *et al.*, 1993) and fruit (Pruvost and Luisetti, 1991). Infection occurs through wounds and, less often, stomata on old leaves (young leaves are thought to be resistant due to their non-functional stomata) (Gagnevin and Pruvost, 2001). High relative humidity (>90%) and moderate temperatures (25–30°C) favour disease development (Kishun and Sohi, 1983; Pruvost and Luisetti, 1991), and disease severity on leaves and fruit is correlated (Manicom, 1986; Pruvost *et al.*, 1990).

Other plants in the *Anacardiaceae* can be infected experimentally, including cashew, imra, mombin and pepper tree (Pruvost *et al.*, 1992; Kishun, 1995). However, strains from these hosts appear to pose little threat to mango (Gagnevin and Pruvost, 2001).

Management

Resistance to bacterial black spot varies greatly among mango cultivars, and resistant cultivars should be used where disease pressure is high (Manicom and Pruvost, 1994). During rainy weather, applications of copper-based bactericides are recommended on susceptible cultivars (Pruvost *et al.*, 1989). Schedules for their application focus on protecting fruit.

When new orchards are established, pathogen-free planting material should be utilized. Since the pathogen moves only short distances in wind-blown aerosols (usually within orchards), the long-distance spread of the pathogen depends almost entirely upon the movement of infected plants (Manicom and Pruvost, 1994). Windbreaks should be used to reduce wounding, and infected twigs should be pruned to reduce inoculum in the canopy.

Black mildew and sooty moulds

Although black mildew and sooty mould are minor problems in well-maintained orchards, the layers of hyphae that these fungi form may be thick enough to block sunlight and reduce photosynthesis. These blemishes also reduce the aesthetic quality and marketability of fruit.

Symptoms

The black mildew and sooty mould fungi produce dark coloured growths on the surfaces of stems, leaves and fruit of mango (Lim and Khoo, 1985). These range from diffuse webs to thick, felty layers of dark hyphae. The variation in symptoms is probably caused by the different species of fungi that are involved.

Causal agents

Black mildew and sooty mould are similar in appearance, but their respective causal agents are distinct (Lim and Khoo, 1985). The black mildew agent, *Meliola mangiferae*, is an obligate parasite. Members of the order *Melioliales* produce dark mycelium with two types of hyphopodia (Alexopoulos *et al.*, 1996). Capitate hyphopodia are lobed appressoria from which infection haustoria are formed, whereas mucronate hyphopodia function as conidiogenous cells. Recent molecular work places these fungi in the pyrenomycetes.

In contrast, the fungi that cause sooty moulds are diverse saprophytes that require honeydew (insect excreta) to colonize plant surfaces. In Malaysia, Lim and Khoo (1985) listed genera in the coelomycetes (*Polychaeton*), hyphomycetes (*Tripospermum*) and loculoascomycetes (*Antennulariella*, *Chaetothyrium*, *Limacinula* and *Scorias*), whereas the reported agents in India were hyphomycetes (*Leptoxylum*, *Microxyphium* and *Tripospermum*) and loculoascomycetes (*Capnodium*) (Butler and Bisby, 1931; Prakash, 1988). Identifying the specific sooty mould agents on affected tissues can be difficult since these fungi may not sporulate and are often found together on a given leaf surface.

Epidemiology and management

Sooty moulds develop on honeydew that is produced on plant surfaces by aphids, mealybugs, scales and other sucking insects. These problems dissipate when the associated insects are controlled with oils or insecticides (Lim and Khoo, 1985).

Blossom blight

Blossom blight can reduce fruit set and production considerably since affected flowers die and large areas of the panicle may be affected. When this disease was controlled with fungicides in the Philippines, a 55–80% increase in fruit set occurred (Pordesimo, 1982).

Symptoms

Blossom blight starts as a wilt of the affected part of the inflorescence that is often curved (Fig. 15.5). The peduncle blackens and dies back from the tip. Internally, discoloration advances beyond the surface lesion. Large black lesions can develop lower on the peduncle, and once it is girdled will kill the apex. Symptoms that were caused by



Fig. 15.5. Symptoms of blossom blight on panicles of mango. Note the blighted, curved terminals and almost complete lack of fruit set (photo: D. Benscher).

Alternaria alternata and *Colletotrichum gloeosporioides* in South Africa were small, mainly superficial black spots, 1–2 × 2–5 mm, on the peduncle (Darvas, 1993; Lonsdale and Kotzé, 1993). Rather than blossom blight, Lonsdale and Kotzé (1993) indicated that these pathogens caused blossom spot.

Causal agents

The cause of blossom blight is somewhat confused and in need of additional research. *C. gloeosporioides* and *C. gloeosporioides* var. *minor* (in Australia) have been reported most frequently as the responsible fungi (Fitzell *et al.*, 1984; Jeffries *et al.*, 1990), although *A. alternata* has also been reported to attack panicles and reduce fruit set (Cronje *et al.*, 1990). Powdery mildew, caused by *Oidium mangiferae*, also damages panicles, but its symptoms are distinct from blossom blight; it is covered separately in this chapter, as are the descriptions of *A. alternata* (Alternaria rot) and *C. gloeosporioides* (Anthracnose).

Recent studies in South Africa indicate that two species of *Fusicoccum* may be important causes of blossom blight. Lonsdale and Kotzé (1993) reported that *F. parvum* (listed as *Dothiorella mangiferae*) caused extensive, systemic damage, and Darvas (1993) indicated that *F. aesculi* (reported as *D. dominicana*) was the only fungus that caused typical symptoms of blossom blight. Descriptions of these fungi appear in Chapters 3 and 1, respectively.

Although *F. aesculi* and *F. parvum* have not been reported as blossom blight agents in other countries, they are associated with mango decline and stem-end rot in several different areas (see below). Work is needed to determine the extent of *Fusicoccum*-incited panicle disease, and what agents are the most important blossom blight pathogens.

Epidemiology

Little is known about the epidemiology of *Fusicoccum*-incited panicle disease. Studies on the stem-end rot diseases have shown that the causal fungi are endophytes. The roles that internal and external sources of inoculum play in the development of panicle disease are not known.

Fitzell *et al.* (1984) investigated environmental conditions that were conducive to infection by *C. gloeosporioides* var. *minor*. They indicated that temperature and free moisture were important determinants of infection, and developed a model that was used to schedule fungicide application. The model reduced the number of applications that were needed to control blossom blight.

Management

Blossom blight must be controlled in order for optimum fruit set and development to occur. Once flowering begins, early and frequent fungicide applications are often needed, especially in areas with high rainfall. Presumably, systemic fungicides would be needed to control *Fusicoccum*-incited disease.

Galls and scaly bark

Gall and scaly bark disorders of mango are known in several producing regions. These diseases are usually minor problems, but can be serious. Although the symptoms vary, the disorders are discussed here under the same heading due to their gross similarities and their probable common cause.

Symptoms

Bark scaling develops as deep cracks along the entire rootstock portion of the plant (Fig. 15.6), and cracks may penetrate the phloem and become necrotic (Prakash and Srivastava, 1987). These symptoms resemble those of a scaly bark disorder, 'cuarteadó', in Colombia (Cook, 1975). In Hawaii and Oahu, similar symptoms developed on mango seedlings (Cook *et al.*, 1971). The bark from the soil line to the first branches was rough and scaly, and xylem pegs, 5 mm long, were evident when the bark was removed around leaf scars and secondary branches.

In Mexico, a disorder known as 'nanahuatē', 'bolas' or 'buba of mango', causes galls, 5–10 cm in diameter, that resemble cauliflower, are initially light green, but become dark brown as they die (Angulo and Villapudua, 1982). The galls remain attached



Fig. 15.6. Bark scaling on the rootstock portion of a mango tree in Israel (photo: R.C. Ploetz).

to trees for many years, and severely affected branches die. Similar symptoms have been noted in Florida, and are associated with pruning injuries (Fig. 15.7). Larger galls have also been noted in Puerto Rico and Florida in the USDA-ARS mango germplasm collection in Miami (Ploetz *et al.*, 1996b; R. Rodriguez, personal communication). The latter galls have been present for >20 years, and range up to 45 cm in diameter (Fig. 15.8). They have rough, scaly exteriors, and usually are found on the main trunk or scaffold limbs of affected trees. Cracks may penetrate the phloem and become necrotic, but the branch death that is associated with galls in Mexico and India has not been observed.

Causal agent

Fusarium decemcellulare causes the above diseases in Florida, Mexico and Venezuela (Malaguti and Reyes, 1964; Angulo and Villapudua, 1982; Ploetz *et al.*, 1996b). Colonies on potato dextrose agar (PDA) are

dark carmine red on the underside. The fungus produces microconidia in false heads or chains on branched and non-branched monophialides (Fig. 15.9). Very large macroconidia ($55\text{--}92 \times 5.5\text{--}7 \mu\text{m}$) are produced in slimy yellow sporodochia, ~1 mm in diameter. The fungus's teleomorph, *Albonectria rigidiuscula* (Rossman *et al.*, 1998), has not been observed on mango.

F. decemcellulare causes corky bark, gall, canker and dieback diseases on diverse woody hosts in the subtropics and tropics (Holliday, 1980; Farr *et al.*, 1989; Alfieri *et al.*, 1994). It causes an important disease of cacao, the cushion gall disease, as well as a stem gall on loquat, and scaly bark of pongam. It has been associated with the corky bark disease of lychee in Florida (see Chapter 14). Host specialization in the fungus has not been reported.

Although *F. decemcellulare* has not been reported to cause gall and scaly bark disorders of mango in other areas, neither have other agents. In the early 1990s, the possible role of *Agrobacterium tumefaciens* was examined in Miami. The bacterium could not be recovered from affected tissues (R. McGuire, personal communication). In Hawaii, microorganisms were not recovered from affected plants, and mechanical transmission tests to common virus indicator plants were negative. Cook (1975) compared the symptoms in Hawaii with those of citrus exocortis, a disease that was thought to be caused by a virus at the time, but is now known to be caused by a viroid. Since viroids were first recognized as plant pathogens at about the same time, Cook *et al.* (1971) did not examine tissue for viroids.

Epidemiology

Isolates of *F. decemcellulare* from mango are only mildly aggressive (Ploetz *et al.*, 1996b). Experimentally, they require wounding in order to infect mango. A recent outbreak of scaly bark in a commercial mango orchard in Florida was associated with pruning wounds (Fig. 15.7).

In the cushion gall disease on cacao, *F. decemcellulare* interacts with several different pest and pathogenic agents, most



Fig. 15.7. Gall, caused by *Fusarium decemcellulare*, that has formed at the site of a previous pruning wound on mango (photo: R.C. Ploetz).

notably capsid bugs (Holliday, 1980). In the corky bark disease of lychee, an armoured scale is associated with the disease (J.E. Peña, personal communication). These insects apparently facilitate infection and disseminate the pathogen. Insect-*F. decemcellulare* interactions have not been investigated on mango.

Management

No pesticides have been identified to control this problem. Measures that should be helpful include the removal and destruction of affected branches and trees in the orchard, disinfestation of pruning equipment to ensure that the pathogen is not spread during pruning operations, and the use of healthy planting material in new orchards.

Internal breakdown

The fruit of many mango cultivars are susceptible to physiological disorders that are referred to, generically, as internal breakdown (Schaffer, 1994). They result from the breakdown of the mesocarp in various locations of the fruit. Three general types of damage are recognized, 'stem-end cavity', 'jelly seed' and 'soft nose'.

These disorders can lead to severe postharvest losses. For example, 70% of the fruit of 'Tommy Atkins' was lost due to internal breakdown in some production areas in Guatemala in 1990 and Nicaragua in 2002 (Schaffer, 1994; R.C. Ploetz, personal observations). Losses in South Florida usually range from 5 to 20%, but can exceed 50% when fruit ripen on the tree.



Fig. 15.8. A large gall on mango cv. 'Langra' in the USDA mango germplasm collection in Miami, USA, caused by *Fusarium decemcellulare*. The machete provides scale (photo: R.C. Ploetz).

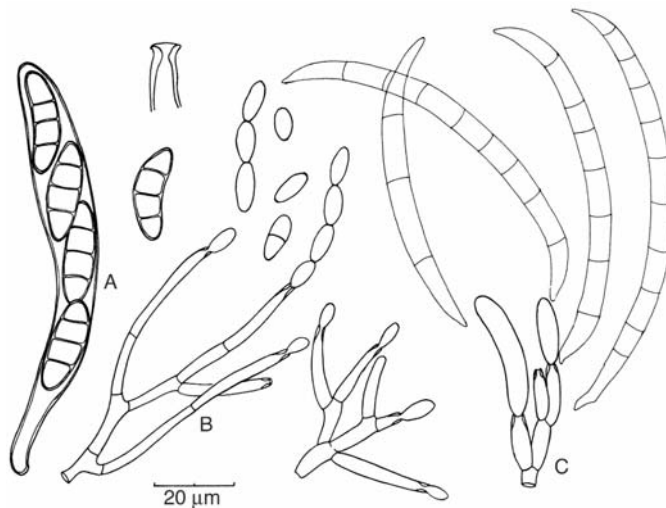


Fig. 15.9. (A) Ascus and ascospores of *Albonectria rigidiuscula*, and (B) microconidia and conidiophores, and (C) macroconidia and conidiophores of its anamorph, *Fusarium decemcellulare* (from CMI description no. 21).

Symptoms

The mesocarp loses physical integrity and dissolves or becomes softened and gelatinous. These problems are most pronounced in ripe fruit (Raymond *et al.*, 1998).

Stem-end cavity begins 8 weeks after fruit set, and eventually becomes an open cavity in which a physical connection between the pedicle and seed is lost (Plate 90) (Raymond *et al.*, 1998). The affected area may be darkened, due possibly to the accumulation of tannins or a phytotoxic buildup of resin (Raymond *et al.*, 1998).

Jelly seed is a general term for the watery breakdown of the mesocarp adjacent to the seed (Fig. 15.10) (Schaffer, 1994). Affected tissue appears to be over-ripe, and the symptoms also begin ~8 weeks after fruit set. Eventually, large areas around the seed can be affected (Raymond *et al.*, 1998). Although these areas usually are water soaked, they can also be spongy and greyish black, not unlike tissue in the stem-end cavity disorder.

Soft nose refers to water soaking of the mesocarp as occurs in the jelly seed disorder (Schaffer, 1994; Raymond *et al.*, 1998). It almost always develops at the distal end of fruit, but can eventually affect larger areas and become darkened (Fig. 15.11).

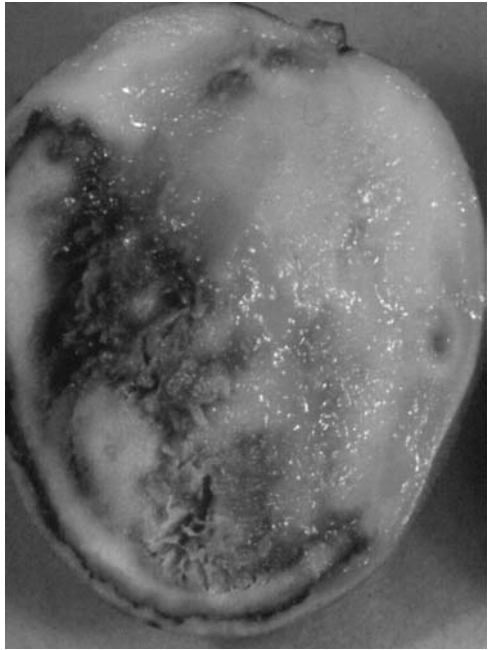


Fig. 15.11. 'Kent' mango fruit with the soft nose disorder (photo: C.W. Campbell).

During a recent episode of internal breakdown on 'Tommy Atkins' in Nicaragua, the overwhelming symptom was of 'pepita negra' (black seed). Externally, the first



Fig. 15.10. Mango fruit with the jelly seed disorder (photo: C.W. Campbell).

symptom of pepita negra is a small black dot at the sinus or beak that internally involves a blackened mesocarp and associated seed (Fig. 15.12). Eventually, classic soft nose symptoms develop on affected fruit. Why symptoms of pepita negra begin in the sinus region, and whether soft nose also begins development in this region are not known. Soft nose and pepita negra always develop on mature fruit.

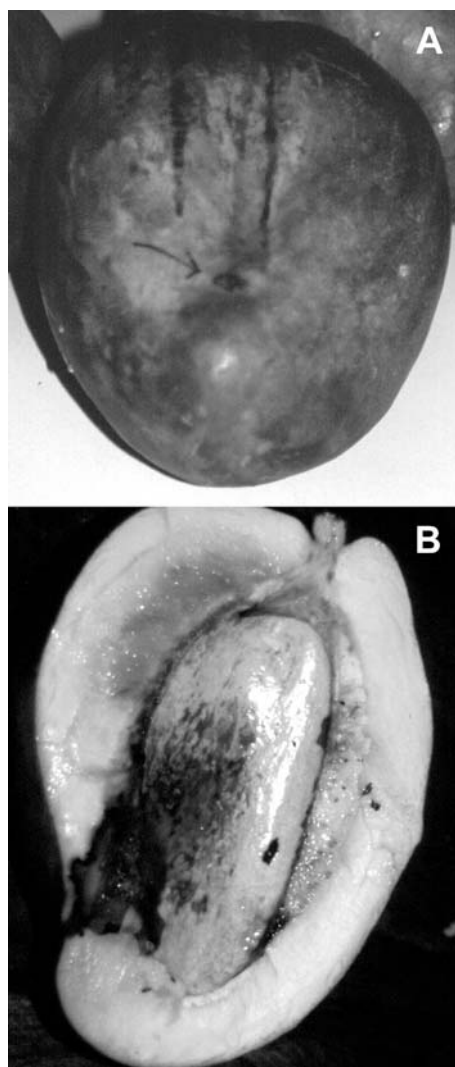


Fig. 15.12. (A) External (arrow points to darkened sinus) and (B) internal symptoms of pepita negra on a 'Tommy Atkins' mango fruit in Nicaragua (photo: R.C. Ploetz).

Cause

The cause of these disorders is somewhat controversial, but several lines of evidence suggest that insufficient levels of calcium in the mesocarp are responsible. That calcium is essential for the integrity and postharvest quality of several different fruits, including mango, is well established (Simmons *et al.*, 1998). Calcium plays a crucial role in preserving cell membrane integrity and strengthening cell walls, and calcium deficiencies in fruit are often associated with increased cell permeability and disintegration.

Low levels of calcium in either mango foliage or fruit have been correlated with internal breakdown (Young, 1957; Young and Miner, 1960; Burdon *et al.*, 1991; Tarmizi *et al.*, 1993). In addition, increased nitrogen fertilization, which would dilute calcium in tissues due to a stimulation of vegetative growth, has also been correlated with an increased incidence of this problem.

Recently, Cracknell (personal communication) completed compelling studies on Tenerife. He applied different levels of calcium and nitrogen to 'Tommy Atkins' to induce different concentrations of these elements in fruit. Breakdown severity was correlated with mesocarp concentrations of nitrogen (positive, $R^2 = 0.95$), calcium (negative, $r^2 = 0.73$) and nitrogen:calcium ratios ($R^2 = 0.93$). When plants received 0.72 mM nitrogen and either 0.62 or 2.5 mM calcium, 75% or more of the fruit were edible (breakdown ratings of 2 or lower). However, when plants received 5.36 mM nitrogen and either 0.62 or 2.5 mM calcium, only 2 and 15% of the fruit, respectively, were edible.

Management

Cultivars with an Indian pedigree are most susceptible to internal breakdown, and those with a fibrous mesocarp are least susceptible (Malo and Campbell, 1978; Schaffer, 1994). The incidence of the disorders is influenced by season and geographic location, and a higher incidence has been reported in sandy versus calcareous soils. On acidic, sandy soils, the disorder has been reduced in fruit of 'Keitt' with soil applications of CaCO_3 or

foliar sprays of $\text{Ca}(\text{NO}_3)_2$ (Young and Miner, 1960). Since high rates of nitrogen fertilization, particularly in the NH_4 form, have increased the prevalence of this problem, excessive levels of this nutrient should not be used.

Malformation

One of the most serious diseases of mango is malformation (Prakash and Srivastava, 1987; Kumar and Beniwal, 1992; Ploetz, 2001). The term 'malformation' refers to the abnormal growth of inflorescences that occurs on affected trees. Since affected inflorescences do not set fruit, the disease can greatly reduce fruit production. Estimated losses range as high as 80% (Ginai, 1965). Vegetative shoots are also affected, and the names 'bunchy top' and 'witches' broom' are used to describe the distortion of these organs in some areas.

Malformation was first reported in India in 1891 (Watt, 1891), and is now recognized in Brazil, Central America, Egypt, Florida

(USA), Israel, Malaysia, Mexico, Pakistan, South Africa, Sudan, Swaziland and Uganda (Crookes and Rijkenberg, 1985; Lim and Khoo, 1985; Kumar and Beniwal, 1992; Ploetz and Prakash, 1997). The disease's greatest impact is in arid production areas.

Symptoms

Young trees in nurseries are most vulnerable to vegetative malformation (Kumar and Beniwal, 1992). Shoots from apical or axillary buds are misshapen and have dramatically shortened internodes (Plate 91). Leaves are dwarfed, and are narrow, brittle and bend back towards the supporting stem. Shoots do not expand fully, resulting in a tightly bunched appearance of these portions of the plant. If all buds on a plant are affected, it remains stunted.

Primary and secondary axes on inflorescences are shortened, thickened and highly branched (Fig. 15.13) (Kumar and Beniwal, 1992; Ploetz and Prakash, 1997). Malformed panicles produce up to three times the normal number of flowers, and these are usually



Fig. 15.13. Symptoms of floral malformation of mango. Note the shortened internodes, large number and size of flowers, and leaves within the inflorescence (phyllody) (photo: R.C. Ploetz).

larger than normal size (Fig. 15.14). Malformation increases the number of male flowers in an inflorescence, and hermaphroditic flowers that are produced are either sterile or, if fertilized, eventually abort. Affected blossoms often produce vegetative structures. Malformed panicles retain a robust, healthy appearance long after non-malformed panicles have set fruit and shed non-fertilized flowers.

Causal agents

The cause of malformation has been controversial for many years. Among the suggested causes have been mites (Narasimhan, 1954), nutritional problems (Prasad *et al.*, 1965), hormonal imbalances (Singh and Dhillon, 1989), viruses (Kausar, 1959), phytoplasmas (Kumar and Beniwal, 1992) and unknown factors (Kumar and Beniwal, 1992).

Convincing evidence that a fungus causes malformation has been in the literature for decades. In India, Summanwar *et al.* (1966) and Varma *et al.* (1974) were the first to report that *Fusarium moniliforme* (recognized later as *F. subglutinans*) caused, respectively, the floral and vegetative forms of the disease. Since the

first reports from India, this fungus has been shown to cause malformation in Egypt, Florida, Israel and South Africa (Ibrahim *et al.*, 1975; Manicom, 1989; Ploetz and Gregory, 1993; Freeman *et al.*, 1999).

Recently, Steenkamp *et al.* (1999, 2000) examined phylogenetic relationships in this pathogen with β -tubulin and histone H3 gene sequences. They indicated that a group of isolates from Florida, India, Israel and South Africa were closely related. They were described later as members of a new species, *F. mangiferae* (Britz *et al.*, 2002). The authors indicated that *F. mangiferae* is conspecific with isolates of '*F. subglutinans*' that previously had been shown to cause mango malformation worldwide. In the same studies, another group of malformation isolates from South Africa was shown to be phylogenetically distinct from the above group. It was described as a second, new species, *F. sterilihyphosum* (Britz *et al.*, 2002). *F. sterilihyphosum* recently has been recovered from malformed mango panicles in Brazil (Zheng and Ploetz, 2002). Although pathogenicity tests have not been performed with isolates of *F. sterilihyphosum*, their recovery from only malformed mango trees suggests that this species also



Fig. 15.14. A comparison of healthy and malformed mango inflorescences (photo: R.C. Ploetz).

causes malformation. *F. mangiferae* and *F. sterilihyphosum* are members of the *Gibberella fujikuroi* species complex, but do not form a *G. fujikuroi* teleomorph (Leslie, 1995; Steenkamp *et al.*, 2000; Ploetz *et al.*, 2002).

Radial growth of *F. mangiferae* on PDA averages 3.4 mm day⁻¹ at 25°C (Britz *et al.*, 2002). Cultures produce aerial, white mycelium, and may be rosy buff or dark purple in reverse. Discrete or confluent sporodochia are cream to orange. Conidiophores are sympodially branched and bear mono- and polyphialides. Microconidia are usually oval, single-celled and 4.3–14.4 (9.0) × 1.7–3.3 (2.4) µm. Macroconidia are long and slender, usually four- to six-celled and 43.1–61.4 (51.8) × 1.9–3.4 (2.3) µm. Chlamydospores are not produced.

On PDA, *F. sterilihyphosum* grows 4.8 mm day⁻¹ at 25°C (Britz *et al.*, 2002). Aerial mycelium is nearly white and the undersides of colonies range from straw to greyish rose to light purple. Sporodochia are cream to orange, but rarely present. Conidiophores are sympodially branched and bear mono- and polyphialides. Microconidia are usually oval, single-celled and 4.5–14.2 (8.8) × 1.6–3.5 (2.6) µm. Macroconidia have a slightly beaked apical cell, a footlike basal cell, and are usually four- to six-celled and 28.4–47.1 (37.1) × 2.4–4.1 (3.2) µm. Sterile, coiled hyphae are formed (Fig. 15.15), but chlamydospores are not.

Two other species of *Fusarium* have been associated with malformation. *Fusarium oxysporum* was reported to cause malformation in Egypt, Mexico and India (Bhatnagar and Beniwal, 1977; Kumar and Beniwal, 1992; Ochoa *et al.*, 1994; El Khoreiby, 1997). Its causal role has not been clearly demonstrated (Ploetz and Prakash, 1997). The D mating population of *Gibberella fujikuroi* (anamorph: *F. proliferatum*) was recovered from malformed mango trees in Malaysia (Leslie, 1995), but pathogenicity tests with the latter isolates were not conducted.

Epidemiology

The epidemiology of malformation is poorly understood. Long-distance spread of the disease relies on the movement of infected nurs-

ery stock, but the means by which within-tree and tree-to-tree spread is accomplished is not known.

Malformation moves slowly in affected orchards (Kumar and Beniwal, 1992). Macro- and microconidia are probable infective propagules since they form on dead, malformed tissues. However, their sensitivity to UV irradiation (Pinkas, personal communication) and the prevalence of malformation in arid countries which have high levels of sunlight confuse this issue. In a study in India, conidia of *F. subglutinans* (probably *F. mangiferae*) were not detected in rotary traps that were placed in an affected mango orchard (Prakash and Srivastava, 1987). Thus, aerial dissemination of conidia of this pathogen may be uncommon.

Artificial inoculation experiments suggest that wounding is required for natural infection. Manicom (1989) was unable to reproduce symptoms unless tissue was wounded prior to inoculation, and artificial induction of the disease by others relied on wounding before inoculation (Summanwar *et al.*, 1966; Ploetz and Gregory, 1993).

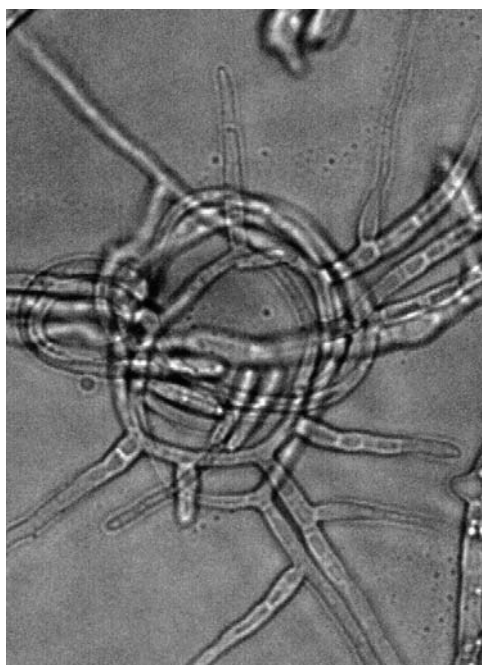


Fig. 15.15. Coiled, sterile hyphae of *Fusarium sterilihyphosum* (photo: Q. Zheng).

Circumstantial evidence suggests that the mango bud mite, *Aceria mangiferae*, is an important factor in the natural development of malformation. It feeds on epidermal cells of floral and vegetative buds of mango, and is often observed in high numbers on malformed trees. Although it has been shown to carry *F. subglutinans* (*F. mangiferae*?) on its body, the pathogenicity of these isolates was not determined (Abdel-Sattar, 1973; Manicom, 1989). Still, it is clear that contaminated mites could play a role in disseminating the fungus and enabling it to infect its mango host (Crookes and Rijkenberg, 1985). Once relationships among *E. mangiferae*, the pathogen and host are understood better, they should provide useful insights into the epidemiology and management of this disease.

F. mangiferae has a highly localized distribution within malformed mango trees (Ploetz, 1994). It predominates in symptomatic tissues, and is found far less frequently, or not all, in asymptomatic organs. This distribution suggests that vegetative and floral buds of mango are the primary sites of infection, and that systemic colonization of older, subtending tissues does not occur. The levels of colonization that occur in affected, but not healthy, tissues appear to indicate that infection thresholds must be reached before symptoms are produced.

Work in South Africa provided good evidence that hormonal imbalances in mango are responsible for the symptoms of malformation (Nicholson and van Staden, 1988; van Staden and Nicholson, 1989; van Staden *et al.*, 1989). Three cytokinins, *trans*-zeatin, dihydrozeatin and ribosyldihydrozeatin, were found in healthy, but not malformed, flowers. In contrast, *iso*-pentenyladenine was detected in malformed, but not healthy, flowers. In culture, the fungus produced *iso*-pentenyladenine derivatives, as well as glucosyl-*O*-zeatin and ribosylzeatin that occur at elevated levels in affected blossoms. The authors suggested that the pathogen produces *iso*-pentenyladenine derivatives in the host that are biologically active and are not found in healthy mango, and also produces glucosyl-*O*-zeatin and ribosylzeatin in the host, resulting in higher than normal levels of these hormones. Furthermore, the fungus

may block production of dihydrozeatin-like compounds in affected tissues. In total, these events would alter the cytokinin:auxin ratio to the extent that the morphology of infected tissues would be altered.

Management

Few approaches to manage malformation are effective (Ploetz and Prakash, 1997). New plantings should be established with pathogen-free nursery stock. Scion material should never be taken from an affected orchard, and affected plants that are observed in the nursery should be removed and destroyed. Nurseries should also not be established in orchards, especially when they are affected by malformation (this practice is common in Egypt and India, two of the most severely affected countries). A new polymerase chain reaction (PCR)-based method for identifying *F. mangiferae* could be useful in preventing the introduction of this pathogen with new germplasm (Zheng and Ploetz, 2002).

Once the disease is found in an orchard, control is possible, but more difficult. Removing symptomatic tissues from trees is usually effective, presumably because it reduces the levels of inoculum in an orchard (Ploetz, 2001). Conventionally, affected terminals and the three subtending nodes are cut from trees, removed from the field and burned. If these measures are practised for 2 or 3 consecutive years, the disease can be reduced to insignificant levels. Thereafter, the disease can be kept in check by removing symptomatic tissues every other year.

Mango decline

Mango decline is a general term that describes several different diseases including blight, canker, gummosis, twig blight, tip dieback and stem bleeding. These diseases are widespread, and can be very destructive (for a thorough review of these diseases, refer to Ploetz and Prakash, 1997). Because these diseases have comparable symptoms and related causal agents, they are treated under the same heading in this chapter.

Symptoms

These problems are not well understood. Symptoms include: marginal necrosis of leaf lamina (Fig. 15.16); defoliation (Fig. 15.16); foliar symptoms of nutritional deficiencies, particularly of iron, manganese and zinc (Fig. 15.16); dieback of small branches from the apex (Fig. 15.16); production of clear or cloudy discharge from terminal buds or cracks in branches and trunks (Plate 92); vascular discoloration; and root degeneration (Ploetz and Prakash, 1997).

Causal agents

Ascomycetes and coelomycetes are associated most often with mango decline (Ploetz and Prakash, 1997), although other biotic or abiotic factors have also been suggested (McSorley *et al.*, 1980; Kadman and Gazit, 1984; Schaffer *et al.*, 1988). The fungal agents are often the same endophytes that cause stem-end rots of mango fruit.

Johnson (1994b) and Johnson *et al.* (1991b, 1992) have shown that several fungi are responsible for stem-end rot, and that host and environmental factors influence the impact of different species. Salient features of these fungi that are listed below are from Johnson (1994b).

Some of the agents have *Botryosphaeria* teleomorphs and *Diplodia* or *Fusicoccum* anamorphs. *D. theobromae* (teleomorph: *B. rhodina*) and *F. aesculi* (teleomorph: *B. dothidea*) are described in Chapter 1. *D. theobromae* causes wilting, defoliation, leaf scorch, gummosis, vascular discoloration and cankers in El Salvador, Egypt, Florida, India, Indonesia, Malaysia and Puerto Rico (Muller, 1940; Das Gupta and Zacchariah, 1945; Alvarez-García and López-García, 1971; Acuña and Waite, 1977; Lim and Khoo, 1985; Ploetz *et al.*, 1996a). Orchard trees and grafted seedlings in the nursery are affected. *F. aesculi* causes twig and branch dieback in Australia, Florida and South Africa (Johnson *et al.*, 1991a; Darvas, 1993; Ploetz *et al.*, 1996a). The same fungus is an important stem-end rot pathogen in Australia.

F. mangiferum causes sudden wilting of shoots and large branches, leaf scorching, trunk cankers and gummosis in Niger



Fig. 15.16. In the foreground, tip dieback of a mango shoot, and in the background zinc-deficient shoot tips (lower) and marginal leaf necrosis (upper) on a tree affected by mango decline. The tree in a desert production area in Israel was water stressed, one of the factors that predisposes mango to this syndrome (photo: R.C. Ploetz).

(Reckhaus and Adamou, 1987). On oatmeal agar (OA), it produces grey, felted mycelium, and on PDA, dark grey mycelium with a 'pepper-spot' pattern of pycnidial initials. Conidia are shorter and broader than those produced by *F. aesculi*, and usually are single-celled, hyaline, ellipsoid to ovoid and $9\text{--}18 \times 4\text{--}6 \mu\text{m}$ (Fig. 15.17); microconidia, $2.5 \times 4.5 \mu\text{m}$, sometimes are observed. Mycelium is composed of branched, brown hyphae that eventually break up into one- or two-celled thallospores (Punithalingham and Waterson, 1970). A *Botryosphaeria* teleomorph develops occasionally on OA.

In Florida, *Alternaria alternata*, *Colletotrichum gloeosporioides*, *Diplodia theobromae*, *F. aesculi* and two *Phomopsis* spp. caused one or more of the above symptoms

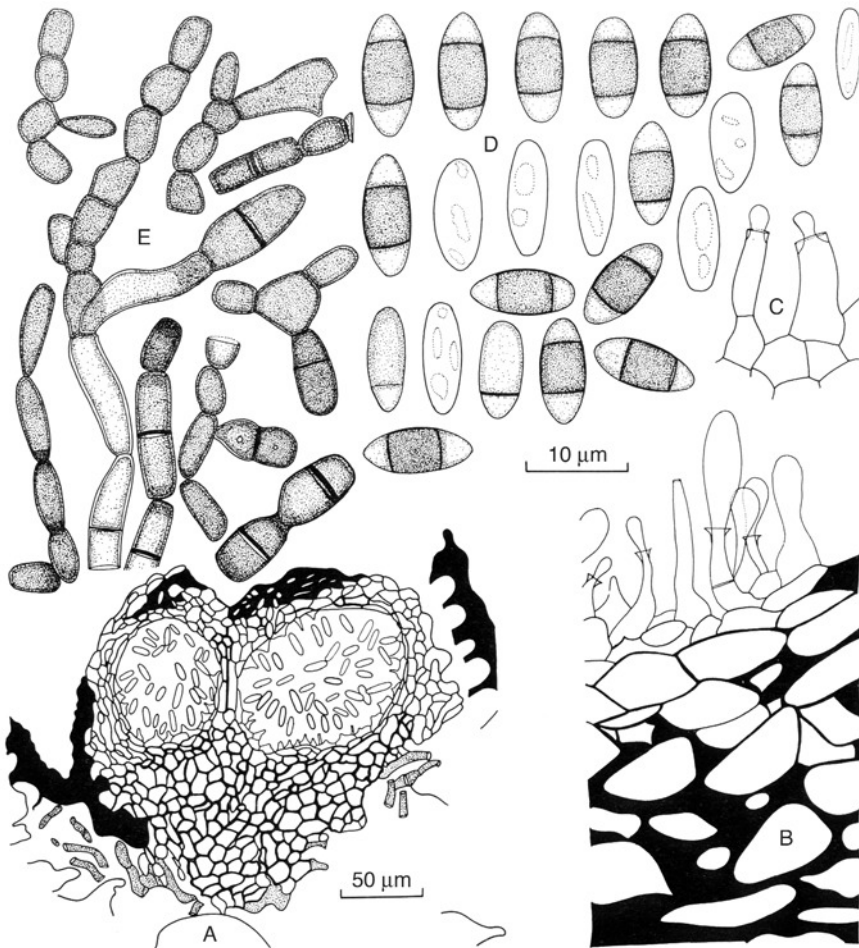


Fig. 15.17. (A) Vertical section of stroma, (B) part of pycnidial wall and conidiophores, (C) conidiophores, (D) conidia and (E) chains of thallospores of *Fusicoccum mangiferum* (from CMI description no. 274).

on 'Keitt' and 'Tommy Atkins' (Ploetz *et al.*, 1996a). *A. alternata* and *C. gloeosporioides* are described in Chapter 1. *C. gloeosporioides*, *D. theobromae* and *F. aesculi* were most virulent, and caused significant necrosis, gummosis and vascular discoloration. Bud necrosis, tip dieback and gummosis symptoms that they caused could not be distinguished unless *C. gloeosporioides* sporulated on inoculated branches.

In Brazil, two species of fungi cause decline disorders. Although the associated symptoms are similar to those that are described above, trees are killed more quickly and frequently than elsewhere.

Diplodia recifensis causes 'Recife sickness' (Batista, 1947). Symptoms include wilting, vascular discoloration, gummosis, blighting of the canopy and tree death. Since *D. recifensis* caused these symptoms only if trees were severely wounded, it was considered a weak pathogen.

An equally destructive disease, called 'seca', 'murcha' or mango blight, is caused by *Ceratocystis fimbriata* (anamorph: *Chalara* sp.) (Viegas, 1960; de Toledo Piza, 1966; Ribiero, 1980). Symptoms are identical to those of Recife sickness, and it is found in many of the same areas. Inoculations with *C. fimbriata* reproduced these symptoms and

verified the tolerance of some cultivars to the disease (Viegas, 1960; Ribiero, 1980). *C. fimbriata* produces brown to black globose perithecia, 140–220 μm in diameter, with long (up to 900 μm), slightly tapered necks (Fig. 15.18) (Morgan-Jones, 1967). Ascospores are elliptical, hyaline, non-septate, smooth, $4.5\text{--}8 \times 2.5\text{--}5.5 \mu\text{m}$, and have a gelatinous sheath that forms a hat-shaped brim. Conidiophores are slender, septate, hyaline to pale brown, and up to 160 μm long. Conidia are cylindrical, hyaline, smooth and $11\text{--}25 (15) \times 4\text{--}5.5 \mu\text{m}$. Chlamydospores are terminal in chains, obovate to oval, thick walled, brown and $9\text{--}18 \times 6\text{--}13 \mu\text{m}$.

Epidemiology

The above fungi are usually weak pathogens. Some are endophytes, and infec-

tion by all is facilitated by wounding. Damage that the fungi cause is greatest in trees that have been weakened by different factors, including water stress; extreme summer and winter temperatures; sun scorch; tar or tanglefoot; high humidity in propagation nurseries; hardpan soils; and nutritional deficiencies, particularly of manganese, iron and zinc (Muller, 1940; Das Gupta and Zacchariah, 1945; Alvarez-García and López-García, 1971; Acuña and Waite, 1977; Schaffer *et al.*, 1988; Ploetz *et al.*, 1996a).

Scolytid beetles are thought to play key roles in the development of Recife sickness and seca in Brazil (Batista, 1947; Viegas, 1960; de Toledo Piza, 1966; Ribiero, 1980). The insects produce galleries in the cambium of affected trees. They feed primarily on fungi, but may act as wounding agents that facilitate infection by, and transmission of, these

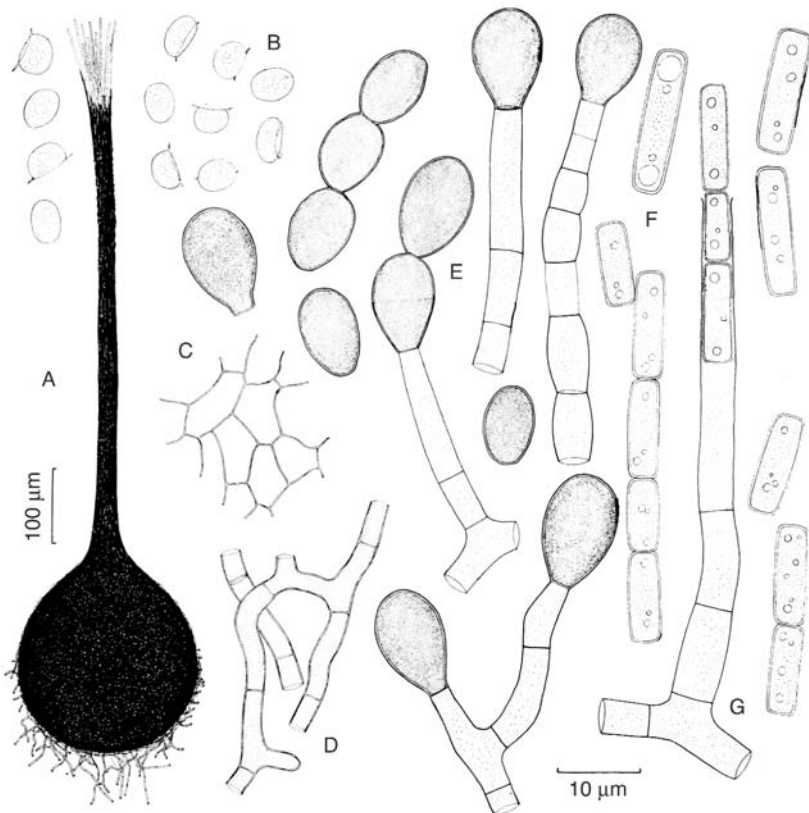


Fig. 15.18. (A) Perithecium, (B) ascospores and (C) surface of the perithecium wall of *Ceratocystis fimbriata*, and (D) mycelium, (E) chlamydospores, (F) conidia and (G) conidiophore of its anamorph, *Chalara* sp. (from CMI description no. 141).

pathogens. Ribiero (1980) reported that *Hypocryphalus mangiferae* was the primary species involved in disseminating *C. fimbriata* since it was the only scolytid found on healthy, as well as diseased, trees.

Management

Managing the decline diseases is difficult since many facets of their epidemiology are not well understood. Reducing the impact of predisposing factors that can be managed, such as drought stress, should be beneficial, but apparently has not been studied. Presently, no protocols exist for either identifying or producing pathogen-free germplasm that could be used to establish decline-free orchards. The internal location and diversity of causal agents also decrease the opportunity for controlling these diseases with fungicides (Peterson *et al.*, 1991). Broad-spectrum, systemic fungicides might be effective, but have not been widely tested. Injecting fungicides, as is done to control Dutch elm disease, might also be effective, but has not been tested. In India, dieback was controlled by removing affected portions of the canopy and treating the wounded areas with a 5:5:50 Bordeaux mix (Prakash and Raoof, 1989).

Phytophthora-incited diseases

Phytophthora palmivora causes diseases of mango in several different areas. It caused wilt, crown rot, root rot and the death of nursery trees in Arizona, the Philippines and Thailand (Kueprakone *et al.*, 1986; Matheron and Matejka, 1988; Tsao *et al.*, 1994). Gumming and conspicuous bark lesions develop aboveground on these plants, whereas root and crown rots are evident at or below the ground level (Fig. 15.19). Crowded conditions, in addition to excessive irrigation and rainfall, exacerbate these diseases. Presumably, sanitation, the use of less crowded conditions and reduced frequency of irrigation would help to control these problems.

Damage has also been recorded on the trunks of field-grown, mature trees in Côte

d'Ivoire (Lourd and Keuli, 1975), and on fruit in Malaysia and West Africa (Turner, 1960; Chee, 1969). Non-confirmed reports of damage on mature trees come from Colima and Yucatan, Mexico and Esquintla, Guatemala (C.W. Campbell and R. Campbell, personal communications). Mortality of trees is not observed, but substantial stem cracking and bleeding do occur. The specific symptoms which occur on fruit have not been recorded.

P. palmivora is described in Chapter 1. Considering its cosmopolitan distribution and wide host range, as well as the diverse symptoms it causes on mango, it is surprising that these diseases have not been reported more widely. Failures to use selective media or specialized techniques that are required to isolate members of this genus may be responsible (Tsao, 1990).



Fig. 15.19. Crown rot symptoms on a potted mango caused by *Phytophthora palmivora*. The bark has been removed from the lower stem to expose the interior of the damaged crown (photo: M.E. Matheron).

Pink disease

Pink disease affects many economically important woody plants in the humid tropics, and is one of the most destructive diseases of mango in these regions (Holliday, 1980). The disease is also known as cobweb, rubellosis and thread blight (Prakash and Srivistava, 1987). It has been studied most thoroughly on rubber, a very susceptible and important host in Southeast Asia (Rao, 1975).

On mango, pink disease can significantly reduce the vigour of trees and fruit yields (Lim and Khoo, 1985). Trees with dense canopies are most susceptible. In Malaysia, the disease is very common in inland areas with heavy rainfall.

Symptoms

Silky, white mycelial threads develop at the forks of branches or twigs (Lim and Khoo, 1985). Under moist conditions, threads coalesce to form a thin, rough pink incrustation on the bark. This stage takes up to several months to develop and coincides with the penetration of bark and internal colonization of the tree. Affected bark often cracks. As the fungus kills the vascular and cambial areas beneath the bark, branches above the colonized areas die, resulting in a sparse, patchy canopy. Basidiospores of the pathogen are formed in a pinkish white hymenial layer on the pink crust. Orange pustules, the pathogen's sporodochia, may also develop on infected bark.

Causal agent

The causal organism is a basidiomycete, *Erythricium salmonicolor* (anamorph: *Necator decretus*) (Holliday, 1980; Lim and Khoo, 1985). It has a wide host range and is described in Chapter 1.

Epidemiology and management

The anamorph and teleomorph of the pathogen are formed during wet conditions and are dispersed by rainsplash and wind. Cultural practices and fungicides are helpful. See Chapter 1 for details.

Powdery mildew

Powdery mildew is a widespread and important disease of leaves, panicles and fruit of mango. The disease can reduce yields by as much as 90%, due mainly to the effect the disease has on fruit set and development (Schoeman *et al.*, 1995).

Symptoms

Mango cultivars vary considerably in their resistance to powdery mildew (Palti *et al.*, 1974). On the most susceptible cultivars, all foliar, floral and fruit parts of the plant are affected (Fig. 15.20). Since affected panicles may set few or no fruit, and because infected small fruit may abort, it is on these portions of the plant that the disease is most important.

Leaves may also suffer considerable damage, and young leaves are most susceptible. A white, powdery coating, the conidia, forms on the underside, both sides or the top side of leaves, depending on the cultivar that is



Fig. 15.20. Powdery mildew on panicles of a resistant (left, 'Glenn') and susceptible (right, 'Nam Doc Mai') mango cultivar (photo: G.I. Johnson).

affected (Fig. 15.21). When damage occurs on the underside of leaves, often it is restricted to the midrib, and in all cases the leaf becomes distorted. These areas become purplish and eventually necrotic. Mature leaves are not susceptible.

On affected panicles, the powdery growth can cover all tissues, resulting eventually in a brown, shrivelled necrosis. Fruit which have set on such panicles may be shed. Skin cracking and corky tissue occurs on fruit that are infected when they are about the size of a pea, and young fruit abscise (Fig. 15.22). As they enlarge past the size of a marble, they are no longer susceptible.

Causal agent

Berthet (1914) first described *Oidium mangiferae* in Brazil, and it is now recognized in most of the important mango-producing regions in the world (Palti *et al.*, 1974). The fungus affects only mango (Prakash and Srivistava, 1987).

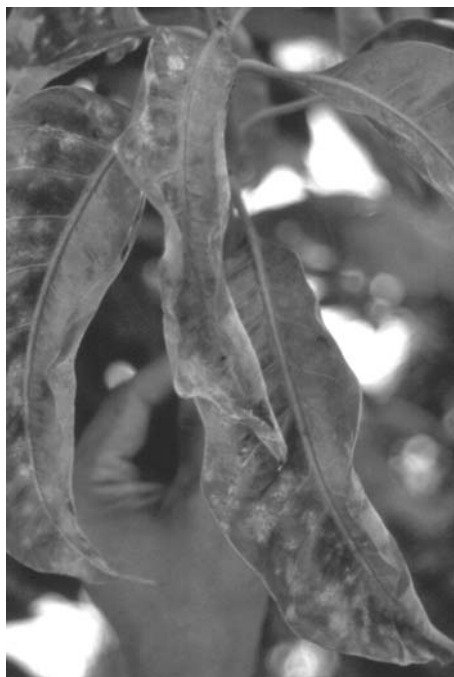


Fig. 15.21. Mango leaves that are affected by powdery mildew. Note the superficial coverage of mycelium of the causal fungus, *Oidium mangiferae*, on the leaves (photo: G.I. Johnson).

Conidium and haustorium traits indicate that *O. mangiferae* belongs to the *Erysiphe polygoni* group (Johnson, 1994a). Although the pathogen originally was classified as *E. cichoracearum* by Wagle (1928), Uppal *et al.* (1941) indicated that it produced saccate and lobed appressoria, which are not characteristic of *E. cichoracearum*.

The pathogen has no known teleomorph, a common trait for fungi that cause powdery mildews in the tropics (Holliday, 1980). Conidia of *O. mangiferae* are single-celled, hyaline, elliptical to barrel shaped, and $33\text{--}43 \times 18\text{--}28 \mu\text{m}$ (Uppal *et al.*, 1941; Palti *et al.*, 1974). They are produced in very large numbers on host surfaces, and impart a powdery appearance to affected tissues that is responsible for the disease's common name. Conidiophores are of the pseudoidium type, with 2–4 septa and a straight basal cell (Boesewinkel, 1980). Germ tubes terminate in appressoria that in turn produce globular haustoria in the host's epidermal cells.

Epidemiology

Powdery mildew is most severe during cool, dry weather (Gupta, 1989). Conidia are wind disseminated and are released on a diurnal basis (Schoeman *et al.*, 1995). Peak spore release, between 1100 and 1600 h, was positively correlated with temperature and negatively correlated with humidity, vapour pressure deficit and leaf wetness (all significant at $P < 0.01$). The cardinal temperatures for conidium germination are 9, 32 and 23°C (Palti *et al.*, 1974). Conidia germinate at a very wide range of relative humidity (as low as 20%).

Infection can take place within 5–7 h, and conidia are produced within 5 days of infection. Disease development occurs between 10 and 31°C, and 60 and 90% relative humidity.

Management

Schoeman *et al.* (1995) noted that symptoms of powdery mildew first appeared 2–3 weeks after 20% of the inflorescences were at the red coloured to red open stage of panicle development. On this basis, they recom-



Fig. 15.22. Powdery mildew on small mango fruit. Note the cracking and conspicuous sporulation of *Oidium mangiferae* on the fruit surface. Once fruit become larger than a pea, they are no longer susceptible (photo: G.I. Johnson).

mended that the first fungicide application should occur when panicles begin to change colour. Assuming an effective period of 3 weeks for a given application, they concluded that applications should continue every 3 weeks after the initial application and until panicle susceptibility decreased at the end of fruit set.

Powdery mildew is easily controlled with sulphur, although its use may burn flowers and young fruit during warm, sunny conditions (Johnson, 1994a). Several newer fungicides are effective, but must be used at the onset of symptom development in order to be effective. Work in Israel demonstrated the efficacy of a sterol-inhibiting fungicide, diniconazole, a strobilurin fungicide, Kresoxymethyl, and a phosphate fertilizer, KH_2PO_4 , over several years (Reuveni *et al.*, 1998). When the fungicides were used either in

alternation or in tank mixes with the fertilizer, enhanced disease control was realized.

Root rot and damping-off

Pythium vexans has been reported to cause root rot and wilt of seedlings (Lim and Khoo, 1985). In Malaysia, this pathogen caused seedling losses of up to 30% in nurseries. Symptoms included wilting of foliage which initially becomes pale green, but later develops necrotic patches. Roots develop a wet, blackened necrosis which begins in fine roots and progresses to larger roots and the root collar. Death of seedlings often occurs. In addition to overcrowding and excessive moisture, Lim and Khoo (1985) indicated that the use of polybags were also factors in the development of this problem.

Prakash and Singh (1980) reported that the basidiomycete *Rhizoctonia solani* (teleomorph: *Thanatephorus cucumeris*) caused root and damping-off of seedlings in India. It is described in Chapter 12. Affected tissues become soft, dark brown or black, and seedlings ultimately may become completely girdled and collapse. Mycelia and sclerotia of the pathogen are formed conspicuously on affected tissues.

Scab

The disease was first recognized in Cuba and Florida in the 1940s. It is now widespread in the western hemisphere, and recently was reported in Australia (Cook, 1975; Alcorn *et al.*, 1999). It is caused by *Denticularia mangiferae*.

Scab is most important in nurseries since young host tissues are most susceptible, and because the moist environments which are often found there aid infection (Ruehle and Ledin, 1955). Lesions, usually first observed on the underside of leaves, are dark brown to black, and 1–2 mm in diameter. They may enlarge to 5 mm in diameter and become light grey with narrow, dark borders. Infected foliage develops a distorted appearance. Greyish blotches are produced on stems.

Although Bitancourt and Jenkins did not prove an associated teleomorph, *Elsinoë mangiferae*, was related to *D. mangiferae* (Bitancourt and Jenkins, 1943; Alcorn *et al.*, 1999), it is described on the assumption that they are related. *E. mangiferae* produces brownish ascocarps, $30\text{--}48 \times 80\text{--}160 \mu\text{m}$, in the host epidermis. Globular asci, $10\text{--}15 \mu\text{m}$ in width, are dispersed in ascocarps, and contain 1–8 hyaline ascospores. Ascospores are $4\text{--}6 \times 10\text{--}13 \mu\text{m}$, four-celled and constricted at the median septum; the subapical cell is longitudinally septate. Conidiophores of *F. mangiferae* are erect, sinuous, $2.5\text{--}3.5 \times 12\text{--}35 \mu\text{m}$, and wider at the base. Conidia are brown, one- or two-celled, and $2\text{--}4 \times 6\text{--}29 \mu\text{m}$.

Host tissues are most susceptible when young, and generally decrease in susceptibility as they mature. Although rainy weather promotes sporulation of the fungus, specific information is not available on the epidemiology of scab. Whether or not conidia and ascospores are infectious is not known.

Stem-end rot

Johnson (1994b) wrote a very thorough description of stem-end rot and its diverse causal agents. Interested readers are referred to this publication for more information than is provided in this chapter.

Stem-end rot is a postharvest disease that increases in importance as orchards become older and when preharvest fungicide programmes reduce the incidence and severity of anthracnose. Losses can increase during prolonged storage at low temperatures, or when fruit ripen at $>28^\circ\text{C}$.

Symptoms

Variable symptoms are caused by different fungi at the stem end upon ripening. *D. theobromae*, *Fusicoccum aesculi* (Plate 93) and *F. mangiferum* (Plate 94) cause diffuse, water-soaked symptoms that spread from the stem end and darken. Necrosis is subcuticular and may affect the entire mesocarp within a week at 25°C . Subsequently, mycelium of the causal fungi may develop on, and tan liquid may drain from, the fruit exterior. *Phomopsis mangiferae* produces dark lesions at the stem end that penetrate the mesocarp, but do not spread as quickly as those caused by the above fungi (Plate 95). These lesions may resemble stem-end anthracnose caused by *Colletotrichum gloeosporioides*, but the latter lesions usually penetrate the mesocarp no deeper than 10–20 mm and are covered with salmon pink spore masses of the pathogen (Fig. 15.2). *Cytosphaera mangiferae* causes slow-spreading, tan lesions at the stem end similar to the symptoms produced by *Aspergillus niger*. *C. mangiferae* produces characteristic conidiomata around the stem end. *Pestalotiopsis mangiferae* causes tan lesions that develop slowly and eventually are covered with acervuli.

Causal agents

Several fungi cause stem-end rot (Johnson, 1994b). *D. theobromae* is most important in hot regions, whereas *F. aesculi* predominates in cooler areas. Less frequent causes of stem-end rot are *Colletotrichum gloeosporioides*, *F.*

mangiferum, *Phomopsis mangiferae*, *Pestalotiopsis mangiferae*, *Cytosphaera mangiferae* and two *Dothiorella* taxa (*Dothiorella* 'long' and *D. aromatica*) that were placed recently in *Fusicoccum* (Slippers *et al.*, 2001). Diagnostic features of *C. gloeosporioides*, *Diplodia theobromae* and *F. aesculi* appear in Chapter 1, and those of *F. mangiferum* are found in this chapter under 'Mango decline'.

Phomopsis mangiferae produces dark brown, subglobose, usually solitary conidiomata that are 1.2 mm wide and 0.75 mm high (Punithalingham, 1993) (Fig. 15.23). They are uni- or multilocular and have no ostiole. Conidiophores are hyaline filiform, simple or branched, septate and 15–75 μm long. Conidia exude white and occasionally yellow tendrils. α -Conidia are hyaline, single-celled, $6\text{--}9 \times 2\text{--}2.5 \mu\text{m}$, and fusoid with a large guttule at each end.

Pestalotiopsis mangiferae produces acervuli on fruit, twigs and leaves that are up to 300 μm in diameter, amphigenous, scattered, globose to lenticular, or ellipsoidal and subepidermal (Fig. 15.24) (Mordue, 1980). At maturity, acervuli rupture the epidermis with a pore through which conidia exude. Conidiophores are hyaline, cylindrical, septate, branched and $5\text{--}15 \times 1.5\text{--}4 \mu\text{m}$, and

conidiogenous cells are holoblastic, indeterminate and smooth. Conidia are fusiform, straight or slightly curved, slightly constricted at the septa, $16\text{--}24$ (19.8) μm long $\times$ $4\text{--}7.5$ (5.6) μm wide. They have three olivaceous brown median cells and a hyaline apical cell that terminates in $1\text{--}4$ (3) $\times$ $3\text{--}25$ (9.8) μm appendages with obtuse ends.

Cytosphaera mangiferae produces colonies on PDA with light grey to buff mycelium, and dark, immersed crenate margins. Pycnidia are produced in concentric bands. Conidia are hyaline, oblong to ovate, aseptate, very thick walled and $16\text{--}26 \times 10\text{--}13 \mu\text{m}$ (Fig. 15.25).

Epidemiology

The causal fungi are endophytes in mango stem tissue (Johnson *et al.*, 1992). They infect the inflorescence early in its development and reach the fruit pedicel several weeks after flowering begins. They then remain quiescent until fruit ripen.

When they occur simultaneously in fruit, *D. theobromae* predominates over *F. aesculi* at high temperatures, whereas *F. aesculi* predominates at $\leq 25^\circ\text{C}$. Although these fungi generally out-compete other stem-end

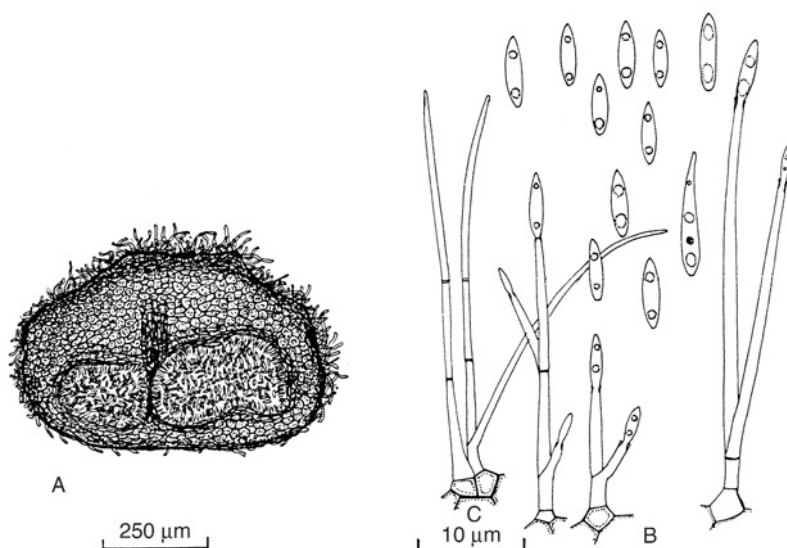


Fig. 15.23. (A) Vertical section of a locular conidiomata, (B) conidiophores, conidiogenous cells and α -conidia, and (C) paraphyses of *Phomopsis mangiferae* (from CMI description no. 1168).

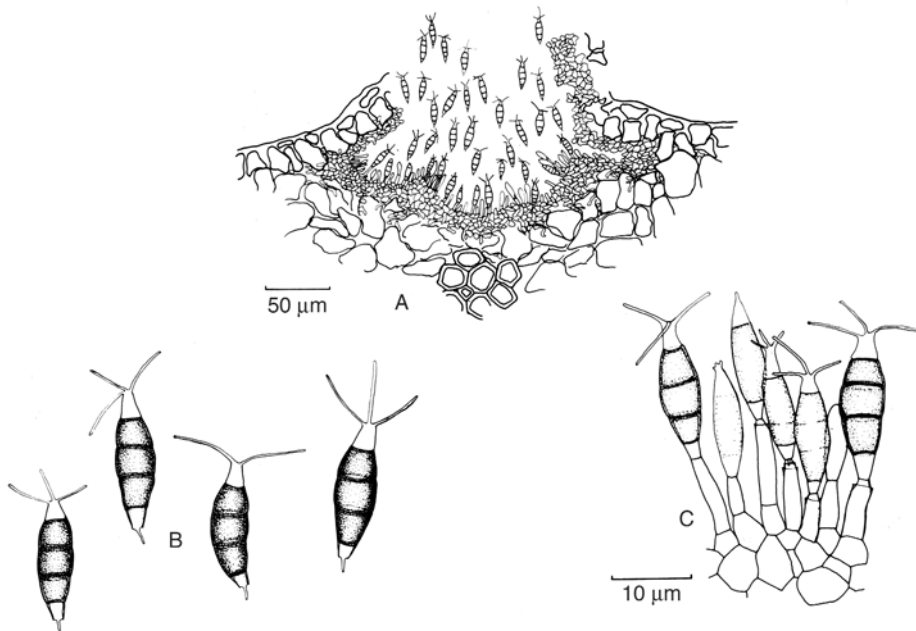


Fig. 15.24. (A) Vertical section of an acervulus, (B) mature conidia and (C) conidiogenous cells and developing conidia of *Pestalotiopsis mangiferae* (from CMI description no. 676).

colonists, *C. gloeosporioides* can repress up to 25% of the *F. aesculi* infections at 13–18°C. Symptoms appear 3–7 days after harvest at 25°C, but can be delayed for an additional 7–12 days at 13°C. Ethylene treatment before storage at 13°C can promote stem-end rot development.

Fruit can also be infected by *D. theobromae* at harvest, particularly when the stem end is inserted in soil to remove sap. These infections develop earlier than endophytic infections. Fruit-to-fruit spread of these pathogens can also occur after harvest.

Management

A dip in hot (52°C) benomyl for 5 min is effective, but not allowed in many areas (Johnson, 1994b). Hot water (55°C) alone or vapour heat treatment are less effective control measures. When fruit are placed in cool storage for long periods, they must be treated with a fungicide. Preharvest applications of copper oxychloride and postharvest application of prochloraz may also reduce stem-end rot, but the level of control

will be lower than that obtained with hot benomyl. Fruit should not be placed in soil for sap removal.

Verticillium wilt

Verticillium wilt of mango was first reported in Florida (Marlatt *et al.*, 1970). Although the cause of the disease was reported originally to be *Verticillium albo-atrum*, this was before *V. dahliae* was widely recognized as a distinct species. Since the causal fungus described by Marlatt *et al.* (1970) formed microsclerotia, it is clear that *V. dahliae* was responsible. *V. dahliae* is described in Chapter 3.

Symptoms of the disease resemble those caused on avocado (see Chapter 3). They include a 'firing' and necrosis of leaves, usually in only a portion of the canopy. The unilateral development of the disease often does not progress to other portions of the tree and, in time, trees may recover. Killed leaves usually remain attached to the tree, and the xylem of affected branches is brown. Internally, the vascular system is brown (Fig. 15.26).

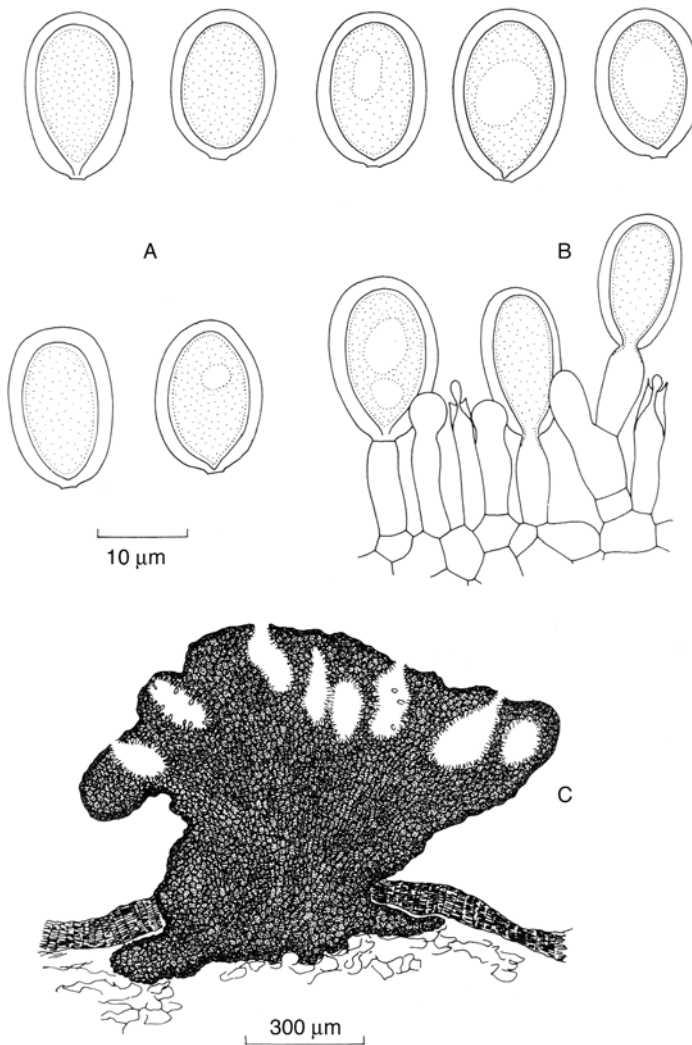


Fig. 15.25. (A) Thick-walled conidia, (B) conidia and conidiophores, and (C) pycnidium of *Cytosphaera mangiferae* (from Sutton, 1980).

Verticillium wilt is relatively uncommon and is usually found on land on which susceptible vegetable crops, such as aubergine and tomato, were grown recently (Pohronezny and Marlatt, 1982). In Egypt, almost all trees that were interplanted with another suspect, potato, were affected by the disease (R.C. Ploetz, unpublished observations). New mango orchards should not be planted on such sites. Alternative control measures for Verticillium wilt are not available. Fumigation with methyl bromide or

other fumigants would not be practical for large sites, and fungicides which are effective against this disease do not exist.

White root disease

Rigidoporus lignosus, the basidiomycete that causes white root disease, is a common soil inhabitant in the humid tropics of Africa and Asia (Holliday, 1980). It has also been reported in the western hemisphere, but



Fig. 15.26. Discoloration of the vascular system in a mango branch caused by *Verticillium* wilt (photo: R.C. Ploetz).

the identity of the fungus in the latter regions is unclear. *R. lignosus* has a large host range on woody perennials that contains many important crops, including rub-

ber, the host on which the pathogen was first reported. The importance of rubber and the very significant losses witnessed by rubber producers in the eastern hemisphere have resulted in a considerable body of research being generated on this pathogen (Nandris *et al.*, 1987). *R. lignosus* is described in Chapter 1.

The most effective means for controlling white root rot rely on eliminating or avoiding colonized woody debris when new orchards are established (Lim and Khoo, 1985). Unfortunately, this is impractical in many situations. Alternative measures include: (i) treating planting holes with elemental sulphur to promote the growth of antagonistic microorganisms; (ii) treating stumps left after clearing operations with chemicals which discourage their colonization by basidiospores; and (iii) establishment of leguminous cover crops which promote growth of the pathogen and the eventual exhaustion of its energy reserves.

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